Cesar Rodriguez

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EDUCATION

University of Michigan **May 2023, Ann Arbor, MI, US**
Bachelor of Science in Engineering in Computer Science - Magna Cum Laude **3.55/4 GPA**
Community Involvement: Michigan Artificial Intelligence Safety Initiative, Society of Hispanic Professional Engineers
Relevant courses:
- Engineering: Machine Learning, Introduction to Artificial Intelligence, Web Systems, Human Centered Software, Introduction to Logic Design, Data Structures and Algorithms, Introduction to Computer Organization, Introduction to Computer Security
- Philosophy: Mind and Machine, Descartes to Kant, Early Modern Philosophy, Critical Reasoning
- Statistics: Statistics and Artificial Intelligence, Computational Methods, Introduction to Statistical Computing

EXPERIENCE

NextArena **Chicago, IL, US**
Chief Executive Officer **July 2026 - Present**
- Lead a team of three engineers and one graphic designer developing an esports tournament platform, launched October 2026.
- Developed an automated outreach and email system with Python and Amazon SES for player announcements and for partnership outreach to content creators, esports lounges and collegiate esports teams.

Full Stack Developer - Technical Lead **March 2024 - July 2026**
- Implemented the company website in accordance with branding specifications, collaborating with designers to deliver an intuitive user experience.
- Developed and documented a React/TypeScript frontend, Django REST APIs and PostgreSQL schema to optimize data flow and performance.
- Built real-time chat and notifications over WebSockets with Django Channels and Redis, and Celery tournament scheduling on Docker and AWS.
- Implemented a payment system supporting cryptocurrency and fiat transactions using PostgreSQL row-level locking for concurrency control, a state machine to enforce valid transaction transitions, and an append-only ledger for auditable transaction history.

MIT Lincoln Laboratory **Westford, MA, US**
Space Systems Analysis and Test Intern - Group 95 **May 2023 - September 2023**
- Automated federally approved RHEL 8 deployments to sensors by building a distributed PXE boot server (TCP, HTTP, TFTP; Bash, Python, C++), cutting deploy time by 95%, and reducing setup complexity.
- Built a hardware-in-the-loop test bench that deployed to a bare-metal machine with simulated sensor input and output, validating OS deployments before they reached the sensors.

Intel Corporation **Folsom, CA, US**
Software Engineering Intern - Cohort Lead **May 2022 - August 2022**
- Created a plugin for an internal tool enabling integration of Intel's new security engine in Intel Firmware Images, improving firmware quality.
- Ensured compatibility with external firmware images by developing Python and Bash tooling that standardized image generation.
- Refactored C++ class hierarchies to improve maintainability and interoperability with the new firmware engine.
- Conducted code reviews and utilized pair programming to develop and implement new features for the new firmware engine.

Intel Corporation **Chicago, IL, US**
Software Engineering Intern - Cohort Lead **May 2021 - August 2021**
- Standardized the stress testing operating system used to validate 14th generation central processor units.
- Earned four Intel recognitions for modernizing legacy Python tooling used across multiple teams.
- Developed and evaluated simulated models to test CPU functionality on FPGA, resulting in a report of findings and recommendations to the team.
- Trained five international team members in Malaysia on usage and implementation of FPGA simulated model for testing CPU functionality.

RESEARCH & FREELANCE

University of Illinois Chicago **Chicago, IL, US**
Freelance Data Analyst **December 2017 - May 2022**
- Extracted unstructured data from image sources to produce human-readable dataset for use by non-technical Museum Studies Program faculty.
- Cleaned, structured, and analyzed datasets supporting research into government accountability.
- Designed data structures to promote ease of use, scalability, and accessibility on a website accessed by tens of thousands of users.

University of Michigan **Ann Arbor, MI, US**
Research Assistant, Cardiovascular Technology **October 2019 - May 2020**
- Interviewed healthcare professionals to identify workflow bottlenecks in emergency cardiovascular response and inform application design.
- Created an application to improve communication between medical personnel that reduced response time from 7 minutes to 5.5 minutes by automating note-taking and streamlining the communication process.

SKILLS

Languages: Python, TypeScript, JavaScript, C++, C, SQL, Java, R, MATLAB, Verilog

Frameworks: React, Django, Django REST Framework, Django Channels, Celery, Zustand, Vite

Technologies & Tools: PostgreSQL, Redis, Docker, AWS (EC2, SES, Secrets Manager), Node.js, Git, Linux, Cadence Perspec

Operating Systems: Windows, Ubuntu, Red Hat Enterprise Linux, macOS